BIOINFORMATICS

Workshop 2: Heuristic Search: Seeding, Banded DP and Log-Odds Scoring

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Learning Objectives

- Build a c-mer index table and use it to find exact-match seeds, as FASTA does.
- Implement banded dynamic programming and a simple ungapped seed-extension step, as a FASTA/BLAST-style search would.
- [Extension task] Estimate a probability model and compute log-odds alignment scores from it.

Background

Banded DP restricts the alignment recurrence to a width-k band of diagonals around a chosen centre, cutting the time from O(mn) to O(cn). FASTA finds a good diagonal by indexing all length-c words of the database sequence; BLAST instead extends short high-scoring (not necessarily exact) seed words. Extension: Separately, the lecture showed that an additive score sigma(a,b) = log[p(a,b)/(q(a)q(b))] is exactly a log-odds test of a 'related' model against a 'random' one.

Coding Activity

1. Write cmer_index(seq, c) returning a dictionary mapping each length-c word to the (1-indexed) list of positions where it occurs in seq.
2. Write find_seeds(index_table, query, c) that slides a c-window along query, looks each word up, and returns a list of (i, j) seed pairs. Then write diagonal_histogram(seeds) that groups seeds by diagonal (i – j) and reports the diagonal(s) with the most support.
3. Write ungapped_extend(s, t, i, j, drop_threshold=3) that, given a seed match at (i, j), extends it in both directions along the diagonal, keeping a running score, stopping (in each direction) once the score falls more than some threshold below its best value so far (say 3 below best).
4. Write banded_align(s, t, k, offset=0) that computes the Workshop 1 global-alignment recurrence, but only for cells with |i - j - offset| <= k // 2. Confirm that for k large enough it reproduces the same score as your Workshop 1 align().
5. Put the pieces together: given a database sequence and a query, build the index, find seeds, extend seeds on most promising diagonals, pick the best-supported diagonal by ungapped extended alignment score, then run banded_align() centred on it. Test on a query that is a mutated substring of a longer database sequence.

Extension Tasks (Optional)

a) Experiment with c = 1, 2, 4 and 6 on the same test data and record how the number of seeds found changes, matching the sensitivity/speed trade-off described in the lecture.

b) Write estimate_background(seqs) and estimate_pair_model(aligned_pairs) to get q(a) and p(a,b) from example data, then log_odds_matrix(p, q) returning sigma(a,b) = log(p(a,b)/(q(a)q(b))). Use it in place of the fixed +1/-1/-2 scheme inside align(), or similarly analyse the protein data and compare it to BLOSUM62.